

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MILU | Context: Mymensingh Environmental Scientist — Cross-Regional South Asian Agricultural Knowledge

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Ground-truth labels originate exclusively from Indian exam portals where 'subject experts ensure the accuracy of the answers' [\[Q22\]](#) and AI4Bharat volunteers conducted manual audits [\[Q87\]](#). No Bangladeshi annotators, agronomists, or environmental scientists are documented in the annotation pool (GAP-6 resolution: 'CONFIRMED ABSENT' [\[WEB-17, WEB-18\]](#)). Approximately 45% of questions arrived pre-tagged from source portals, with the remainder receiving GPT-4O-MINI-generated topic assignments [\[Q33, Q34\]](#). There is no documentation of inter-annotator agreement metrics or annotator demographics beyond AI4Bharat affiliation. The expert elicitation (A3) noted Indian exam answers are 'generally considered authoritative' for Bangladeshi environmental scientists, with notable exceptions around trans-border water agreements where Indian answers encode nationally partial perspectives — and the YAML documents that bilateral volatility (Ganges 2026 expiry, Teesta unresolved) makes this gap more acute [\[WEB-11, WEB-14\]](#). Dataset analysis confirms Bangladesh appears only as wrong-answer distractor [\[DATASET-D21, D25\]](#) — Bangladeshi perspectives are structurally excluded. For a HIGH-priority dimension, this is a substantial convergent-validity concern.

ID	Evidence
Q22	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q33	"Upon examination, we found that approximately 45% of questions were accurately labeled with a topic name, while the remaining questions lacked this information." (p.4)
Q34	"To address this issue, we first translate the untagged questions into English using INDICTRANS2 (Gala et al., 2023) and then prompt GPT-4O-MINI model to assign an appropriate topic name to the question." (p.4)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
D21	ଡକ୍ଟ୍ରିନେର 2019 ର 6 ଭାରତୀୟ ଫାର୍ମାକୋପୋଇଆ କ ଦ୍ଵାରା ପ୍ରଥମେ ଜଣା ଦିଆଯାଇଥିବା ଦେଶ କେଉଁ?... ଅଫଗାନିସ୍ତାନ; option2: ବାଙ୍ଗଳାଦେଶ — Which country first reported Indian Pharmacopoeia (Afghanistan) — Bangladesh appears as wrong answer option
D25	డిసెంబర్ 2019లో భారతీయ ఫାର୍మాకోପీయా ను గుర్తించిన మొదటి దేశం ఏదీ?... ఆఫ్ఘనిస్తాన్; option2: బంగ్లాదేశ్ — Same Indian Pharmacopoeia question in Telugu — Bangladesh again used only as distractor
WEB-11	Ganges treaty in renegotiation 2024–2026 — Indian and Bangladeshi positions diverge (https://thediplomat.com/2026/04/why-the-india-bangladesh-ganges-treaty-renewal-must-deliver-real-security/)
WEB-14	No formal Teesta treaty; positions categorically differ (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)
WEB-17	AI4Bharat/MILU GitHub: no Bangladeshi annotator participation (https://github.com/AI4Bharat/MILU)
WEB-18	NAACL paper: annotation by AI4Bharat volunteers and Indian exam-portal experts only (https://aclanthology.org/2025.naacl-long.507/)

Strengths

- Indian exam-portal source provides documented subject-expert validation pathway for ~45% of pre-tagged questions [\[Q22, Q33\]](#), offering some baseline authority for non-contested STEM facts
- Manual cluster review for subject label assignment [\[Q37\]](#) and AI4Bharat volunteer audits [\[Q87\]](#) provide a layer of human oversight

- For non-contested scientific facts (basic chemistry, biology, physics), expert elicitation (A3) confirms Indian exam answers are generally accepted as authoritative by Bangladeshi environmental scientists

ID	Evidence
Q22	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q33	"Upon examination, we found that approximately 45% of questions were accurately labeled with a topic name, while the remaining questions lacked this information." (p.4)
Q37	"We manually review these clusters and assign appropriate subject labels." (p.4)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground truth labels reflect Indian exam-portal subject experts and AI4Bharat volunteers [[Q22](#), [Q87](#)]. For non-contested STEM facts, this is broadly acceptable to Bangladeshi scientists (elicitation A3). For trans-border policy, governance, and Bangladesh-specific agricultural practice, Indian answers encode partial perspectives [[WEB-11](#), [WEB-14](#)].

OC-2: Likely disagreement zones: (a) trans-border river-flow allocation answers, (b) agricultural-policy authority questions where Indian and Bangladeshi national positions differ, (c) Bangladesh-specific cropping-calendar questions (none currently exist in MILU). Elicitation A3 explicitly identified this. Currently, no such questions are present in sampled data, so the disagreement is latent rather than active.

OC-3: Annotator demographics documented are: Indian exam-portal subject experts [[Q22](#)], AI4Bharat team volunteers [[Q87](#)], and GPT-4O-MINI for topic-tag assignment [[Q34](#)]. No Bangladeshi annotators documented [[WEB-17](#), [WEB-18](#)]. No inter-annotator agreement metrics, no demographic breakdown beyond institutional affiliation. INSUFFICIENT DOCUMENTATION on annotator country/region distribution.

OC-4: Re-annotation by a Bangladeshi annotator pool would be required for any deployment-relevant subset, particularly for environmental policy and trans-border water questions. The deployment YAML notes BAU and BARI as candidate institutional sources [[WEB-7](#), [WEB-8](#)].

OC-5: Aggregation method is single-correct-answer MCQ, which by construction erases minority perspectives — a Bangladeshi alternative answer to a contested water-policy question would be marked wrong. Authors do not discuss aggregation alternatives or pluralistic labeling.

OC-6: Documented label issues: (a) no Bangladeshi annotator participation [[WEB-17](#)], (b) no IAA reported, (c) GPT-4O-MINI used for ~55% of topic tagging [[Q34](#)] without human verification documented per item, (d) Bangladesh appears only as distractor [[DATASET-D21](#), [D25](#)]. These violate convergent validity (labels don't correlate with regional perspectives where they would diverge) and external validity for the deployment.

ID	Evidence
Q22	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)

Q34	"To address this issue, we first translate the untagged questions into English using INDICTRANS2 (Gala et al., 2023) and then prompt GPT-4O-MINI model to assign an appropriate topic name to the question." (p.4)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
D21	ଡସିମେବର 2019 ରେ ଭାରତୀୟ ଫାର୍ମାକୋପିଆ ଦ୍ଵାରା ପ୍ରଥମେ ଜଣା ଦିଆଯାଇଥିବା ଦେଶ କେଉଁଟି?... ଅଫଗାନିସ୍ତାନ; option2: ବାଙ୍ଗଳାଦେଶ — Which country first reported Indian Pharmacopoeia (Afghanistan) — Bangladesh appears as wrong answer option
D25	డిసెంబర్ 2019లో భారతీୟ ఫାର୍ମାକୋପିయా ను గుర్తించిన మొదటి దేశం ఏదీ?... ఆఫ్ఘనిస్థాన్; option2: బంగ్లాదేశ్ — Same Indian Pharmacopoeia question in Telugu — Bangladesh again used only as distractor
WEB-7	BAU as professional institutional context — text infrastructure supported (https://bau.edu.bd/)
WEB-8	https://en.wikipedia.org/wiki/Bangladesh_Agricultural_University
WEB-11	Ganges treaty in renegotiation 2024–2026 — Indian and Bangladeshi positions diverge (https://thediplomat.com/2026/04/why-the-india-bangladesh-ganges-treaty-renewal-must-deliver-real-security/)
WEB-14	No formal Teesta treaty; positions categorically differ (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)
WEB-17	AI4Bharat/MILU GitHub: no Bangladeshi annotator participation (https://github.com/AI4Bharat/MILU)
WEB-18	NAACL paper: annotation by AI4Bharat volunteers and Indian exam-portal experts only (https://aclanthology.org/2025.naacl-long.507/)

Information Gaps

- No inter-annotator agreement metrics reported anywhere in MILU documentation
- No demographic breakdown of AI4Bharat volunteer auditors (gender, regional origin, professional discipline)
- Per-item verification rate for GPT-4O-MINI-generated topic tags [Q34]

Requires Expert Verification

- Bangladeshi environmental scientist re-annotation of any cross-border water or Bangladesh-specific agricultural items, were such items added
- Identification by a Bangladeshi policy expert of specific MILU items where Indian exam answers would be considered nationally partial vs. universally authoritative

Remediation

Gap 1: No Bangladeshi annotators; no inter-annotator agreement metrics; ~55% of topic tags assigned by GPT-4O-MINI without per-item human verification

Recommendation: Recruit a Bangladeshi annotator panel including BAU agronomists and an environmental-policy expert to re-annotate any items relevant to Bangladesh, trans-border water, or wetland systems; report inter-annotator agreement at minimum for the Bangladesh-relevant subset.